



FinTrackr

FinTrackr is a full-stack personal finance tracking application that helps users manage accounts, track income and expenses, and receive **weekly financial summaries via Telegram**.

Built with **React, Node.js, Express, MongoDB**, and integrated with the **Telegram Bot API**.



Features



Authentication

- User signup & login using JWT
- Protected routes for authenticated users only



Account Management

- Create, edit, and delete accounts (Bank / Cash / Card / Other)
- Maintain real-time account balances



Transaction Management

- Add income and expense transactions
- Edit and delete transactions
- Automatic balance updates per transaction
- Validation for insufficient balance on expenses



Insights

- Total balance calculation
- Income & expense summaries
- Date-wise transaction history



Telegram Integration

- Telegram bot integration
 - Users can link their Telegram ID
 - **Automated weekly income & expense report**
 - Scheduled using cron jobs (every Monday at 9 AM IST)
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Tech Stack

Frontend

- React (Vite)

- React Router
- Axios
- Tailwind CSS

Backend

- Node.js
- Express.js
- MongoDB + Mongoose
- JWT Authentication
- node-cron
- Telegram Bot API

Deployment

- Render (backend + frontend)
- MongoDB Atlas

Project Structure

```
FinTrackr/  
|  
├── backend/  
│   ├── controllers/  
│   ├── routes/  
│   ├── models/  
│   ├── middleware/  
│   ├── utils/  
│   ├── server.js  
│   └── .env  
├── frontend/  
│   └── my-react-app/  
│       ├── src/  
│       ├── dist/           (production build)  
│       └── package.json  
└── README.md
```

Environment Variables

Create a `.env` file inside the **backend** folder:

```
PORT=5000
MONGO_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret
TELEGRAM_BOT_TOKEN=your_telegram_bot_token
```

⚠️ Do not commit `.env` to GitHub

Running Locally

Clone the repository

```
git clone https://github.com/your-username/FinTrackr.git
cd FinTrackr
```

Install backend dependencies

```
cd backend
npm install
```

Install frontend dependencies

```
cd ../frontend/my-react-app
npm install
```

Build frontend

```
npm run build
```

Start backend server

```
cd ../../backend
node server.js
```

Open  `http://localhost:5000`

Telegram Setup

1. Open Telegram and search for your bot
2. Send `/start` to the bot
3. Copy the **Telegram ID** shown by the bot
4. Paste the ID in the **Telegram Setup** section of FinTrackr
5. Save it

 You will receive a **weekly summary every Monday at 9 AM IST**

Cron Job (Weekly Report)

```
cron.schedule(  
  "0 9 * * 1",  
  async () => {  
    await sendWeeklyReport();  
  },  
  {  
    timezone: "Asia/Kolkata",  
  }  
);
```

Security Notes

- JWT stored in localStorage
 - Protected backend routes using middleware
 - MongoDB queries scoped per user
 - Telegram ID linked securely to user profile
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Future Enhancements

- Monthly & yearly reports
 - Category-wise analytics
 - Email notifications
 - Export transactions (CSV / PDF)
 - Multi-currency support
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Author

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